III B.TECH – I SEMESTER

Chemistry Of Energy Systems

Open Elective-I | JNTUA R23 · B.Tech CSE

Course Code	Title of the Subject	L	T	P	C
	CHEMISTRY OF ENERGY SYSTEMS	3		-	3

COURSE OBJECTIVES
1
2
3
4
5

COURSE OUTCOMES
CO1
CO2
CO3
CO4
CO5

Mapping between Course Outcomes and Programme Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2												
CO3												
CO4												
CO5												

UNIT-1: Electrochemical Systems: Galvanic cell, Nernst equation, standard electrode potential, application of EMF, electrical double layer, polarization, Batteries- Introduction ,Lead-acid ,Nickel- cadmium, Lithium ion batteries and their applications**.**

UNIT-2: Fuel Cells: Fuel cell- Introduction, Basic design of fuel cell, working principle, Classification of fuel cells, Polymer electrolyte membrane (PEM) fuel cells, Solid-oxide fuel cells (SOFC), Fuel cell efficiency and applications.

UNIT-3: Photo and Photo electrochemical Conversions: Photochemical cells Introduction and applications of photochemical reactions, specificity of photo electrochemical cell, advantage of photoelectron catalytic conversions and their applications.

UNIT-4: Solar Energy: Introduction and prospects, photovoltaic (PV) technology, concentrated solar power (CSP), Solar cells and applications. .

UNIT-5: Hydrogen Storage: Hydrogen storage and delivery: State-of-the art, Established technologies, Chemical and Physical methods of hydrogen storage, Compressed gas storage, Liquid hydrogen storage, Other storage methods, Hydrogen storage in metal hydrides, metal organic frameworks (MOF), Metal oxide porous structures, hydrogel , and Organic hydrogen carriers.

Text books

- Physical chemistry by Ira N. Levine
- Essentials of Physical Chemistry, Bahl and Bahl and Tuli.
- Inorganic Chemistry, Silver and Atkins

Reference Books:

** **1. Fuel Cell Hand Book 7th Edition, by US Department of Energy (EG&G technical services

And corporation)

2. Hand book of solar energy and applications by ArvindTiwari and Shyam.
3. Solar energy fundamental, technology and systems by Klaus Jagar et.al.

4.Hydrogen storage by Levine Klebonoff